

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	21/08/2026	Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: _____

Annexure A – Test Case Design Assignment

Project Title: AI-Based Learning Management System with Personalized Learning Recommendations

Domain: Education

1. Problem Analysis

1.1 Problem Statement

Traditional Learning Management Systems mainly provide common learning materials and courses to all students without considering their individual learning abilities, performance, preferences, and progress. This may make it difficult for students to identify suitable courses and learning resources.

The proposed **AI-Based Learning Management System with Personalized Learning Recommendations** uses student learning data, assessment performance, course progress, and learning preferences to recommend suitable courses and learning materials.

1.2 Project Objectives

- To provide secure student registration and login.
- To allow students to browse, search, and enroll in courses.
- To provide access to learning materials and assessments.
- To automatically calculate quiz and assessment scores.
- To track student learning progress.

1.3 Functional Requirements

Requirement ID	Functional Requirement
FR01	The system shall allow students to register an account.
FR02	The system shall allow registered students to log in securely.
FR03	The system shall allow students to browse and search available courses.
FR04	The system shall allow students to enroll in available courses.
FR05	The system shall allow enrolled students to access learning materials.
FR06	The system shall provide quizzes and assessments.
FR07	The system shall automatically calculate assessment scores.
FR08	The system shall track and display student learning progress.
FR09	The system shall generate AI-based personalized recommendations.
FR10	The system shall send notifications related to learning activities.
FR11	The system shall provide secure logout and session management.

1.4 Non-Functional Requirements

Requirement ID	Non-Functional Requirement
NFR01	The system should provide acceptable response time.
NFR02	The system should support multiple users simultaneously.
NFR03	Student information and account data should be protected.
NFR04	The system should remain reliable during normal operation.
NFR05	The system should handle network and server failures appropriately.
NFR06	The user interface should be easy to understand and use.
NFR07	AI recommendations should be relevant to the student's learning profile.
NFR08	The system should maintain data consistency and accuracy.

1.5 Expected Outcomes

The system is expected to provide secure access to educational resources, accurate assessment results, reliable progress tracking, and relevant personalized recommendations. It should provide a responsive and user-friendly learning environment while protecting student information.

2. Test Requirements

2.1 Functional Requirements to be Tested

The following functional requirements will be tested:

- Student registration
- Student login
- Course browsing and searching
- Course enrollment
- Learning material access
- Quiz and assessment functionality
- Score calculation
- Progress tracking

2.2 Non-Functional Requirements to be Tested

- Performance
- Security
- Reliability
- Usability
- Availability

3. Test Scenarios

Scenario ID	Test Scenario
TS01	Verify student registration with valid information.
TS02	Verify registration with invalid and duplicate information.
TS03	Verify student login using valid and invalid credentials.
TS04	Verify course browsing and searching.
TS05	Verify course enrollment.
TS06	Verify access to learning materials.
TS07	Verify quiz and assessment functionality.
TS08	Verify score calculation.
TS09	Verify learning progress tracking.
TS10	Verify AI-based personalized recommendations.
TS11	Verify recommendations based on student performance.
TS12	Verify recommendations based on learning preferences.
TS13	Verify notifications.
TS14	Verify logout and session management.
TS15	Verify unauthorized access prevention.
TS16	Verify network/server failure handling.
TS17	Verify system performance with multiple users.

4. Detailed Test Cases

Test Case	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail Stat
T C01	Student registration with valid	1. Open registration page. 2. Enter valid details. 3. Click Register.	Name: Mohammed, Email:	Student account should be created successfully.	Not Executed	Not Executed
T C02	Registration with invalid	1. Open registration page. 2. Enter invalid email. 3. Submit form.	Email: student@	Appropriate email validation message should be displayed and	Not Executed	Not Executed
T C03	Registration with duplicate	1. Register using an existing email. 2. Submit form.	Existing email: student@test.com	System should reject duplicate registration and display an appropriate	Not Executed	Not Executed
T C04	Student login with valid	1. Open login page. 2. Enter valid username/email and password. 3.	student@test.com / Student@123	Student should be successfully logged in and redirected to the dashboard.	Not Executed	Not Executed
T C05	Login with invalid password	1. Open login page. 2. Enter valid email and incorrect password. 3.	student@test.com / Wrong@123	Login should fail and an appropriate error message should be displayed.	Not Executed	Not Executed
T C06	Login with empty fields	1. Open login page. 2. Leave fields empty. 3. Click Login.	Email: Empty, Password: Empty	Required-field validation messages should be displayed.	Not Executed	Not Executed
T C07	Course browsing and search	1. Login. 2. Open Courses. 3. Search for a course.	Search: Python Programming	Relevant courses matching the search keyword should be displayed.	Not Executed	Not Executed
T C08	Course search with no	1. Open course search. 2. Enter a nonexistent course name.	Search: QuantumXYZ	System should display a suitable “No courses found” message.	Not Executed	Not Executed
T C09	Course enrollment	1. Login. 2. Select an available course. 3. Click Enroll.	Course: Python Programming	Student should be successfully enrolled in the selected course.	Not Executed	Not Executed
T C10	Duplicate course enrollment	1. Enroll in a course. 2. Attempt to enroll in the same course again.	Existing enrolled course	System should prevent duplicate enrollment.	Not Executed	Not Executed
T C11	Learning material access	1. Login. 2. Open enrolled course. 3. Select learning material.	PDF/video/lesson	Selected learning material should open successfully for the enrolled student.	Not Executed	Not Executed
T C12	Unauthorized learning material	1. Login without enrollment. 2. Attempt to access restricted course	Unenrolled course	System should deny access and request course enrollment.	Not Executed	Not Executed

T C1 3	Quiz submission with valid	1. Open quiz. 2. Answer all questions. 3. Submit quiz.	10 valid answers	Quiz should be submitted successfully and evaluated.	Not Executed	Not Executed
T C1 4	Quiz submission with	1. Open quiz. 2. Leave some questions unanswered. 3. Submit.	7 answered, 3 unanswered	System should either allow submission according to requirements or warn about	Not Executed	Not Executed
T C1 5	Quiz score at minimum boundary	1. Complete quiz with all answers incorrect. 2. Submit quiz.	Score: 0/100	System should correctly display 0% without calculation errors.	Not Executed	Not Executed
T C1 6	Quiz score at maximum	1. Complete quiz with all answers correct. 2. Submit quiz.	Score: 100/100	System should correctly display 100%.	Not Executed	Not Executed
T C1 7	Score calculation accuracy	1. Submit quiz. 2. Compare correct answers with displayed	8 correct out of 10	System should calculate and display 80%.	Not Executed	Not Executed
T C1 8	Progress tracking	1. Open enrolled course. 2. Complete lessons. 3. View progress.	5 of 10 lessons completed	Progress should be displayed as 50%.	Not Executed	Not Executed
T C1 9	Progress boundary validation	1. Start a course. 2. Check progress. 3. Complete all lessons.	Start: 0%, Complete: 100%	Progress should correctly change from 0% to 100%.	Not Executed	Not Executed
T C2 0	AI personalized	1. Login as student. 2. View recommendation section. 3. Review	Learning history + interests + progress	System should generate recommendations relevant to the student's profile.	Not Executed	Not Executed
T C2 1	Performance-based recommend	1. Complete assessments with low scores in a topic. 2. Open	Mathematics score: 40%	System should recommend suitable beginner/revision material for the weak topic.	Not Executed	Not Executed
T C2 2	High-performance	1. Complete assessments with high scores. 2. View recommendations.	Python score: 95%	System should recommend more advanced or challenging Python-related	Not Executed	Not Executed
T C2 3	Learning-preference recommend	1. Set preferred learning type. 2. View recommendations.	Preference: Video-based learning	Recommendations should prioritize suitable video-based resources where	Not Executed	Not Executed
T C2 4	Recommendation with insufficient learning	1. Create/login to a new account. 2. View recommendations before completing courses.	New student, no history	System should provide suitable introductory/popular recommendations without generating	Not Executed	Not Executed
T C2 5	Learning notification	1. Enroll in course. 2. Trigger a relevant notification event. 3.	New course/quiz notification	Appropriate notification should be generated and displayed to the student.	Not Executed	Not Executed
T C2 6	Logout functionality	1. Login. 2. Click Logout. 3. Try accessing dashboard using browser	Valid logged-in session	Student should be logged out and protected pages should not remain	Not Executed	Not Executed

T C2 7	Unauthorized access to protected	1. Logout. 2. Directly enter a protected dashboard URL.	Protected URL	System should redirect the user to login or display an access-denied message.	Not Executed	Not Executed
T C2 8	Network failure during quiz	1. Start a quiz. 2. Disconnect network during submission. 3.	Network unavailable	System should handle the failure gracefully and provide an appropriate retry/	Not Executed	Not Executed
T C2 9	Server failure during	1. Attempt to open a course while server is unavailable.	Server unavailable	System should display a suitable error message and should not crash or display	Not Executed	Not Executed
T C3 0	Multiple-user performance	1. Simulate multiple students accessing the system simultaneously.	Multiple concurrent users	System should remain available and provide acceptable response time	Not Executed	Not Executed

5. Test Design Techniques

5.1 Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups so that representative values can be tested.

Quiz Score

Partition	Test Data	Expected Result
Valid	0–100	Score accepted
Invalid	Below 0	Input rejected
Invalid	Above 100	Input rejected
Invalid	Alphabetic value	Input rejected

Course Search

Partition	Example	Expected Result
Valid keyword	Python	Matching courses displayed
Valid keyword	Mathematics	Matching courses displayed
Invalid/non-existing keyword	XYZABC	No matching courses message
Empty input	Blank	Appropriate validation/default result

Student Information

Partition	Example	Expected Result
Valid	Proper name/email/password	Accepted
Invalid email	student@	Rejected
Empty field	Blank	Validation message
Duplicate email	Existing email	Registration rejected

5.2 Boundary Value Analysis

Boundary Value Analysis tests values at the minimum, maximum, and values immediately around the boundaries.

Quiz Score Boundary

Value	Expected Result
-1	Reject
0	Accept
1	Accept
99	Accept
100	Accept
101	Reject

Course Progress Boundary

Value	Expected Result
-1%	Reject
0%	Valid
1%	Valid
99%	Valid
100%	Valid
101%	Reject

Boundary cases are covered primarily by **TC15, TC16, and TC19**.

5.3 Decision Table Testing

Decision Table Testing is applied to AI-based personalized recommendations.

Learning History	Performance	Preference	Course Completion	Expected Recommendation
Available	Low	Video	Low	Beginner/revision video material
Available	High	Video	High	Advanced video-based content
Available	Low	Text	Low	Beginner/revision text material
Available	High	Text	High	Advanced text-based content
Not Available	No score	Any	None	Introductory/popular courses
Available	Low	Any	High	Revision/weak-topic resources

This ensures that the recommendation engine considers multiple student factors instead of relying on a single parameter.

5.4 State Transition Testing

The student's learning journey can be represented through the following states:

Logged Out → Logged In → Course Selected → Enrolled → Learning → Quiz → Result → Recommendation → Logout

Current State	Action	Next State
Logged Out	Enter valid credentials	Logged In
Logged In	Select course	Course Selected
Course Selected	Enroll	Enrolled
Enrolled	Open material	Learning
Learning	Start quiz	Quiz
Quiz	Submit assessment	Result
Result	View recommendations	Recommendation
Recommendation	Logout	Logged Out
Logged Out	Access protected page	Access Denied/ Login

6. Requirement Traceability Matrix

Requirement ID	Requirement	Test Case IDs	Coverage
FR01	Student Registration	TC01–TC03	Covered
FR02	Student Login	TC04–TC06	Covered
FR03	Course Browsing/Search	TC07–TC08	Covered
FR04	Course Enrollment	TC09–TC10	Covered
FR05	Learning Material Access	TC11–TC12	Covered
FR06	Assessment/Quiz	TC13–TC16	Covered
FR07	Score Calculation	TC15–TC17	Covered
FR08	Progress Tracking	TC18–TC19	Covered
FR09	AI Personalized Recommendations	TC20–TC24	Covered
FR10	Notifications	TC25	Covered
FR11	Logout/Session Management	TC26–TC27	Covered
NFR01	Performance	TC30	Covered
NFR02	Multiple-user Support	TC30	Covered
NFR03	Security	TC03, TC05, TC12, TC26, TC27	Covered
NFR04	Reliability	TC28–TC29	Covered
NFR05	Failure Handling	TC28–TC29	Covered
NFR06	Usability	TC02, TC06, TC08, TC25, TC28–TC29	Covered
NFR07	AI Recommendation Relevance	TC20–TC24	Covered
NFR08	Data Accuracy	TC17–TC19	Covered

7. Test Coverage

7.1 Functional Requirement Coverage

All identified functional requirements from **FR01 to FR11** have at least one corresponding test case.

Coverage: 100%

7.2 Non-Functional Requirement Coverage

Performance, security, reliability, usability, AI relevance, and data accuracy are included in the test cases.

Coverage: 100% of identified NFR categories

7.3 Normal-Condition Coverage

Normal operations such as registration, login, course enrollment, material access, quiz submission, progress tracking, recommendations, and logout are tested through TC01, TC04, TC07, TC09, TC11, TC13, TC18, TC20, TC25, and TC26.

7.4 Boundary-Condition Coverage

Boundary conditions include:

- Quiz score: **0–100**
- Course progress: **0–100%**
- Minimum and maximum valid values
- Values immediately outside valid ranges

Covered by **TC15, TC16, and TC19** and the Boundary Value Analysis section.

7.5 Invalid-Condition Coverage

Invalid email, duplicate registration, incorrect password, empty login fields, invalid searches, duplicate enrollment, and unauthorized access are covered by **TC02, TC03, TC05, TC06, TC08, TC10, TC12, and TC27**.

7.6 Exceptional-Condition Coverage

Network failure, server failure, incomplete quiz submission, and insufficient recommendation history are covered by **TC14, TC24, TC28, and TC29**.

7.7 AI Recommendation Coverage

AI recommendation testing includes:

- General personalized recommendations
- Low-performance recommendations
- High-performance recommendations

7.8 Security Coverage

- Duplicate account prevention
- Invalid authentication
- Unauthorized learning-material access
- Secure logout
- Protected-page access
- Session management

7.9 Reliability Coverage

Network interruption and server failure scenarios are included to verify that the system handles exceptional situations without data corruption or application crashes.

7.10 Usability Coverage

Validation messages, search results, notifications, error handling, and user navigation are considered in the test cases.

7.11 Testing Technique Coverage

Test Design Techniques	Application	Test Cases
Equivalence Partitioning	Scores, search, registration data	TC02, TC03, TC07, TC08, TC15–TC17
Boundary Value Analysis	Score and progress limits	TC15, TC16, TC19
Decision Table Testing	AI recommendations	TC20–TC24
State Transition Testing	Student learning flow	TC04, TC09, TC11, TC13, TC17, TC20, TC26, TC27

8. Conclusion

The test case design for the **AI-Based Learning Management System with Personalized Learning Recommendations** provides comprehensive coverage of the system's functional and non-functional requirements. The 30 test cases cover normal, boundary, invalid, and exceptional conditions, along with AI recommendation accuracy, security, reliability, performance, and usability.

Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing are applied to improve test coverage and identify potential defects. The Requirement Traceability Matrix ensures that each identified requirement is mapped to appropriate test cases.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/ non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/ scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/ results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/ formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25